

LAB ASSIGNMENT – 14.2

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BATCH – 29

Question:

Task Description 1 – (Landing Page Design)

- Create a simple landing page for a product or service. Include a header, hero section, and footer.

Requirements:

- Use HTML and CSS to structure the page.
- Use AI to suggest layout alignment and typography.
- Ensure the page is responsive across devices.

Task Description 2 – (Theme Toggle Feature)

- Provide a button to switch between light and dark themes..

Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

Task Description 3 – (User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.

Task Description - 4 (Image Gallery with JavaScript)

- Task: Display a small image gallery on a web page.

Requirements:

- Use JavaScript to dynamically load images.
- Add next/previous navigation buttons.
- Use AI to optimize layout responsiveness.

Task Description - 5: (Notification Banner)

- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

PROMPT: Generate a modern responsive web application UI using HTML, CSS, and JavaScript with a

navigation bar, hero section, feedback form, image gallery, and notification banner. Include dark

mode toggle, smooth UI interactions, and clean card-based layout for better user experience.

Combined Code and Output:

Code:

```
from flask import Flask, render_template_string
```

```
app = Flask(__name__)
```

```
@app.route('/')
```

```
def home():
```

```
    return render_template_string("""
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>

<title>Web Lab App</title>
```

```
<style>

body{

    font-family: Arial;

    margin: 0;

    background: #f4f4f4;

}
```

```
/* HEADER */

header {

    background: #333;

    color: white;

    padding: 15px;

    text-align: center;

}
```

```
/* BUTTON */

button {

    padding: 8px;

    margin: 5px;

    cursor: pointer;

}
```

```
/* HERO */

.hero {

    text-align: center;
```

```
padding: 40px;
background: #ddd;
}
```

```
/* CARD */
```

```
.card {
    background: white;
    margin: 20px;
    padding: 20px;
    border-radius: 10px;
}
```

```
/* FOOTER */
```

```
footer {
    background: #333;
    color: white;
    text-align: center;
    padding: 10px;
}
```

```
/* DARK MODE */
```

```
.dark {
    background: black;
    color: white;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- Banner -->
```

```
<div id="banner" style="background:orange;padding:10px;text-align:center;">
```

```
🔥 Welcome Offer!
```

```
<button onclick="closeBanner()">X</button>
```

```
</div>
```

```
<!-- Header -->
```

```
<header>
```

```
<h2>My Web App</h2>
```

```
<button onclick="toggleTheme()">Toggle Theme</button>
```

```
</header>
```

```
<!-- Hero -->
```

```
<div class="hero">
```

```
<h1>Welcome</h1>
```

```
<p>All Tasks Working</p>
```

```
</div>
```

```
<!-- Form -->
```

```
<div class="card">
```

```
<h3>Feedback Form</h3>
```

```
<form onsubmit="return validateForm()">
```

```
<input type="text" id="name" placeholder="Username"><br><br>
```

```
<input type="number" id="rating" placeholder="Rating"><br><br>
```

```
<textarea id="comments" placeholder="Comments"></textarea><br><br>
```

```
<button type="submit">Submit</button>
```

```
<p id="msg"></p>
```

```
</form>
```

```
</div>
```

```
<!-- Gallery -->
```

```
<div class="card">
```

```
<h3>Image Gallery</h3>
```

```
<img id="img" width="250"><br><br>
```

```
<button onclick="prev()">Prev</button>
```

```
<button onclick="next()">Next</button>
```

```
</div>
```

```
<!-- Footer -->
```

```
<footer>
```

```
© 2026 Web Lab
```

```
</footer>
```

```
<script>
```

```
/* THEME */
```

```
function toggleTheme(){
```

```
    document.body.classList.toggle("dark");
```

```
}
```

```
/* FORM */
```

```
function validateForm(){
```

```
    let name=document.getElementById("name").value;
```

```
let rating=document.getElementById("rating").value;
```

```
if(name=="" || rating==""){
```

```
    document.getElementById("msg").innerText="Fill all fields!";
```

```
    return false;
```

```
}
```

```
document.getElementById("msg").innerText="Submitted successfully!";
```

```
return false;
```

```
}
```

```
/* GALLERY */
```

```
let images=[
```

```
"https://via.placeholder.com/250",
```

```
"https://via.placeholder.com/250/ff0000",
```

```
"https://via.placeholder.com/250/00ff00"
```

```
];
```

```
let i=0;
```

```
function show(){
```

```
    document.getElementById("img").src=images[i];
```

```
}
```

```
function next(){
```

```
    i=(i+1)%images.length;
```

```
    show();
```

```
}
```

```
function prev(){
```

```
    i=(i-1+images.length)%images.length;
    show();
}
show();
```

```
/* BANNER */
```

```
function closeBanner(){
    document.getElementById("banner").style.display="none";
}
```

```
</script>
```

```
</body>
```

```
</html>
```

```
""")
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```



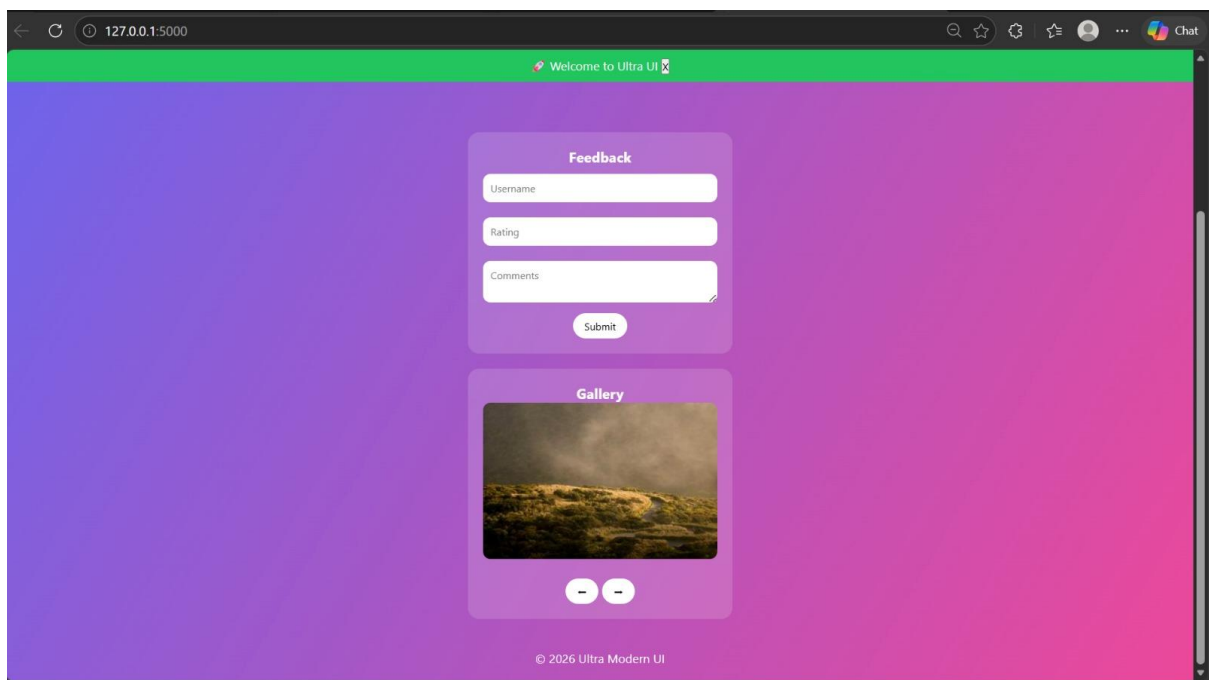
```
147
148 <footer>
149   © 2026 Ultra Modern UI
150 </footer>
151
152 <script>
153
154   /* THEME */
155   function toggleTheme(){
156     if(document.body.style.background.includes("6366f1")){
157       document.body.style.background="black";
158     } else {
159       document.body.style.background="linear-gradient(120deg,#6366f1,#ec4899)";
160     }
161   }
162
163   /* FORM */
164   function validateForm(){
165     let name=document.getElementById("name").value;
166     let rating=document.getElementById("rating").value;
167
168     if(name==" || rating==" ){
169       document.getElementById("msg").innerText="⚠ Fill all fields!";
170       return false;
171     }
172
173     document.getElementById("msg").innerText="✅ Submitted!";
174     return false;
175   }
176
177   /* GALLERY */
178   let images=[
179     "https://picsum.photos/300/200?random=1",
180     "https://picsum.photos/300/200?random=2"
181   ]
182
183   </script>
184 </body>
185 </html>
```

PROBLEMS OUTPUT TERMINAL PORTS

DEBUG CONSOLE

...
ead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit

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JUSTIFICATION :

The design follows a responsive and modern UI structure using cards and flexible layouts.

JavaScript enhances interactivity through theme toggle, form validation, and gallery navigation.

CSS styling improves user experience with clean typography, spacing, and visual hierarchy.

The application demonstrates real-world usability by combining multiple UI components in one page.